

Project Summary: Riva

Riva is an open-source, deterministic performance diagnostics companion for Unreal Engine. It bridges the gap between raw data collection in Unreal Insights and actionable bottleneck resolution for game developers and CI/CD automation pipelines.

The Problem & Solution

Unreal Insights records millions of runtime trace events per minute, but game teams face severe manual overhead scrubbing timelines, sorting through cascading thread symptoms, and catching regressions late in development.

Riva shifts optimization left by providing automated, deterministic diagnostics:

- **Deterministic Spike Detection:** Isolates hitch frames against a rolling median window.
- **Causal Graph Resolution:** Links concurrent stalls, elevating root causes (STUT_STREAMING_IO, STUT_GC) over symptomatic downstream waits (STUT_RHI_SYNC, STUT_CPU_GT).
- **Multi-Spike Correlation:** Clusters repetitive temporal hitches into clean composite findings.
- **Unreal Editor Slate Panel** (SRivaPanel): Dockable tab with bi-directional Unreal Insights timeline selection sync, clipboard actions, and report export dialogs.
- **Headless CI/CD Gatekeeper** (riva): Pure C++20 CLI with zero Unreal Engine header dependencies for instant budget checks (riva check-budget) and regression diffing (riva compare).

Technical Comparison Matrix

Capability	Raw Profilers (Unreal Insights, PIX)	Cloud APM (Sentry, Datadog)	Riva Diagnostics
Primary Output	Raw timeline tracks & millions of events	Stack traces & cloud web dashboards	Actionable, root-cause Markdown/JSON reports
Diagnostic Effort	Manual timeline scrubbing by developers	Aggregated telemetry graphs	100% automated spike detection & bottleneck classification
Symptom Filtering	Shows all thread stalls equally	Shows raw call stacks	Causal Graphing: Elevates root causes (IO/GC) over symptom waits (RHI Sync)
Repetitive Spikes	Spams 50+ identical timeline markers	Aggregates event counts	Multi-Spike Correlation: Clusters repetitive hitches into composite findings
UE Subsystem Context	Raw markers (GarbageCollect, ShaderCompile)	Generic CPU/GPU execution time	Built-in UE5 Signatures: Understands PSO misses, Zen IO, Lumen/VSM GPU passes
In-Editor UI Sync	Separate external window	Web browser overlay	Native Dockable Slate Tab with bi-directional Insights selection linkage
CI/CD Regression Diffing	None (requires human inspection)	Post-release cloud analytics	Headless CLI (riva compare) for automated PR gatekeeping in 5ms
Infrastructure	Local	Required Cloud Accounts & SaaS Fees	100% Local & Offline, zero accounts, zero cloud dependencies

Key Advantages

Riva is **100% open source (MIT License)**, production-ready with a **100% passing 13-executable CTest suite**, and runs **100% locally with zero cloud or account friction**. Its decoupled C++20 core architecture ensures maximum stability across Unreal Engine version updates.